NAME: NIRUTHIKA SRI PRAKASH

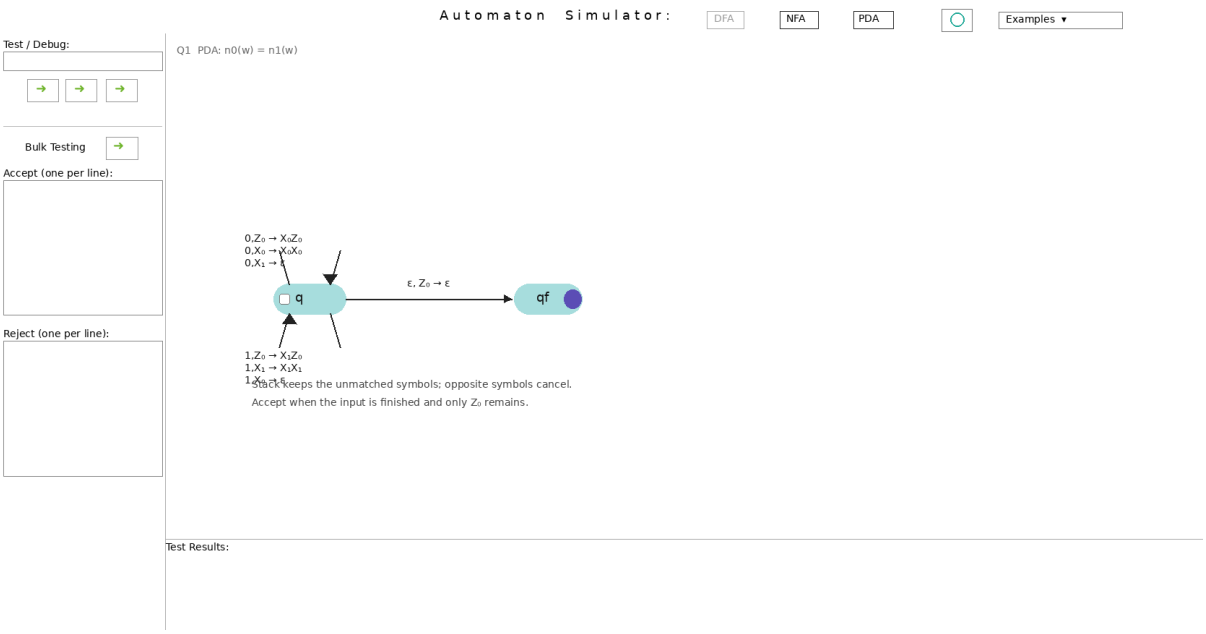
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COURSE CODE: CSA1309

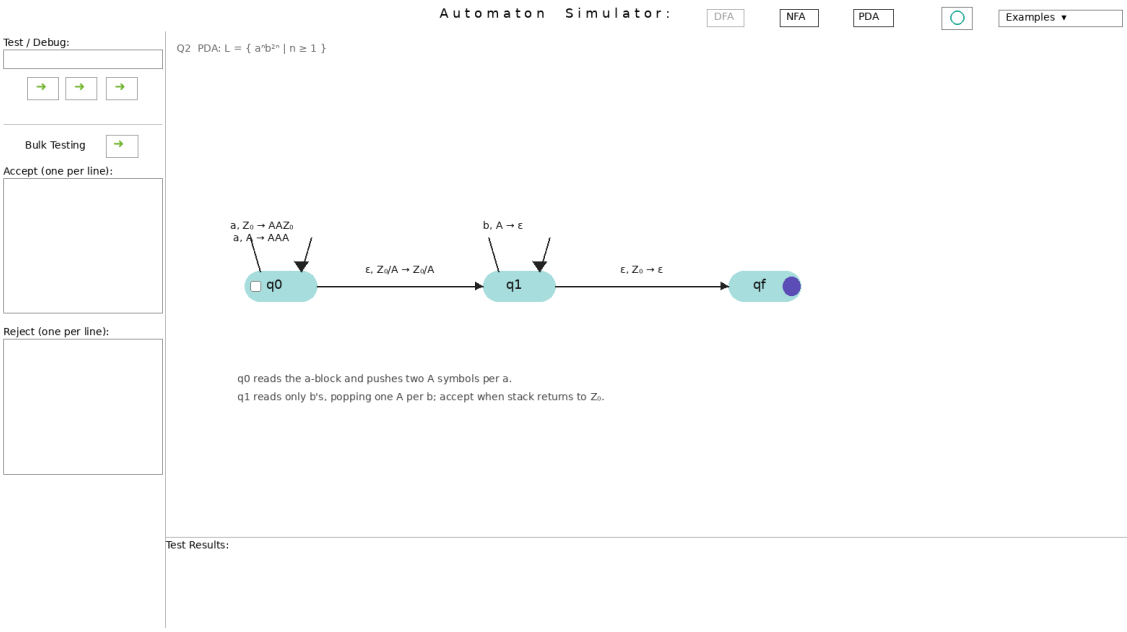
COURSE: THEORY OF COMPUTATION

ASSIGNMENT 4

1. Design a Push Down Automata that accepts the language
   $L = \{w \mid w \in (0+1) * and\ n_0(w) = n_1(w)\ \}$
   $n_0(w)$ is the number of 0's in w
   $n_1(w)$ is the number of 1's in w



2. Design a Pushdown Automata for the language representing a's followed by b's and the number of b's must be twice than that of a's.
   $L = \{a^n b^{2n} \mid n \geq 1\}$

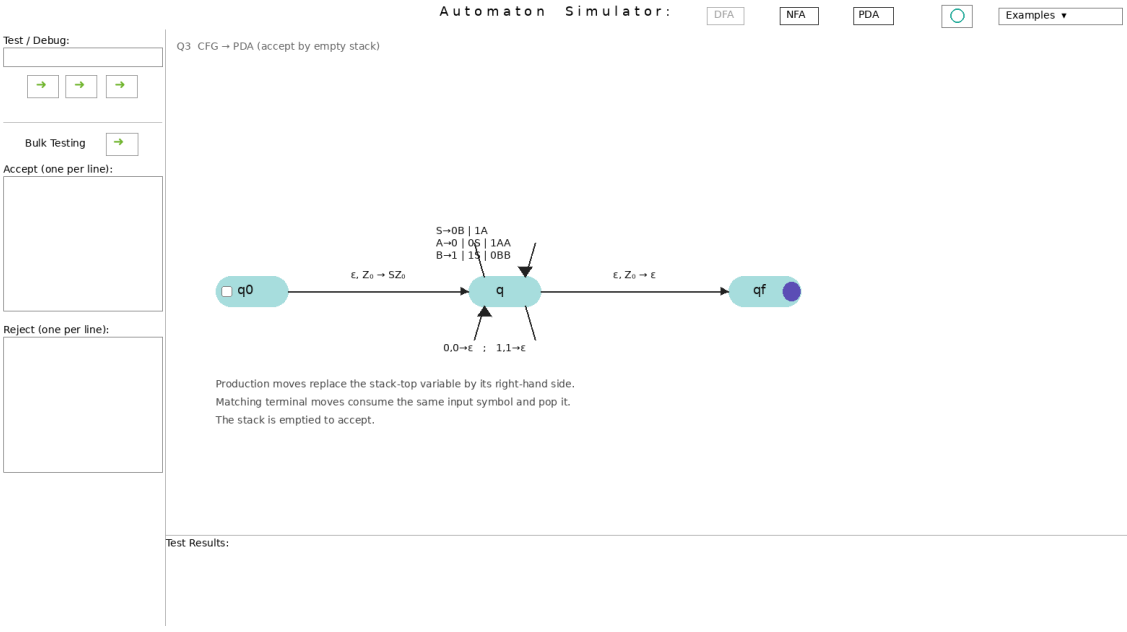


3 Convert the CFG to a Pushdown Automata that accepts by empty stack. Write the formal definition of the PDA.

S → 0B | 1A

A → 0 | 0S | 1AA

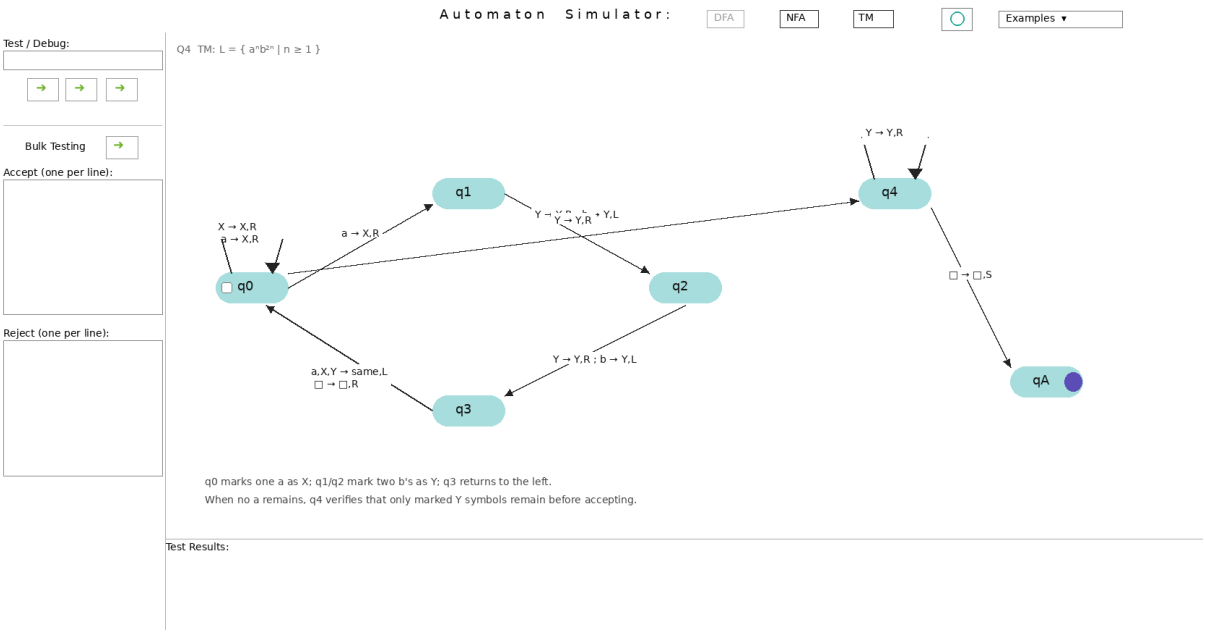
B → 1 | 1S | 0BB



4.Construct a Turing Machine to recognize the language

$$L = \{a^n b^{2n} \mid n \geq 1\}$$

Write the logic used for the design and the formal definition.



5.Construct a Turing Machine to recognize the language

$$L = \{a^n b^n c^n \mid n \geq 1\}$$

Write the logic used for the design and the formal definition.

